

DATA266 Lab 1 — Task 1 Results

Executive summary

This project implements a character-level GPT language model from scratch using PyTorch and TinyStories.

The model predicts the next character using causal multi-head self-attention. No pretrained language model or pretrained embeddings were used.

Seed: 8844 | Dataset: roneneldan/TinyStories

Dataset and Configuration

Dataset configuration

Training sequences: 100000

Validation sequences: 10000

Context length: 128

Epochs: 10

Batch size: 64

Learning rate: 0.0003

Dropout: 0.1

Transformer layers: 4

Attention heads: 4

Embedding dimension: 256

The 100K/10K interpretation is exactly 100,000 training sequences and 10,000 validation sequences. Each sequence has 128 input characters and a shifted target sequence.

Model Architecture

Decoder-only character-level GPT

The embedding layer converts characters into vector representations.

Causal multi-head self-attention allows each character to attend only to current and previous characters.

Four Transformer blocks process the sequence before the final linear layer predicts the next character.

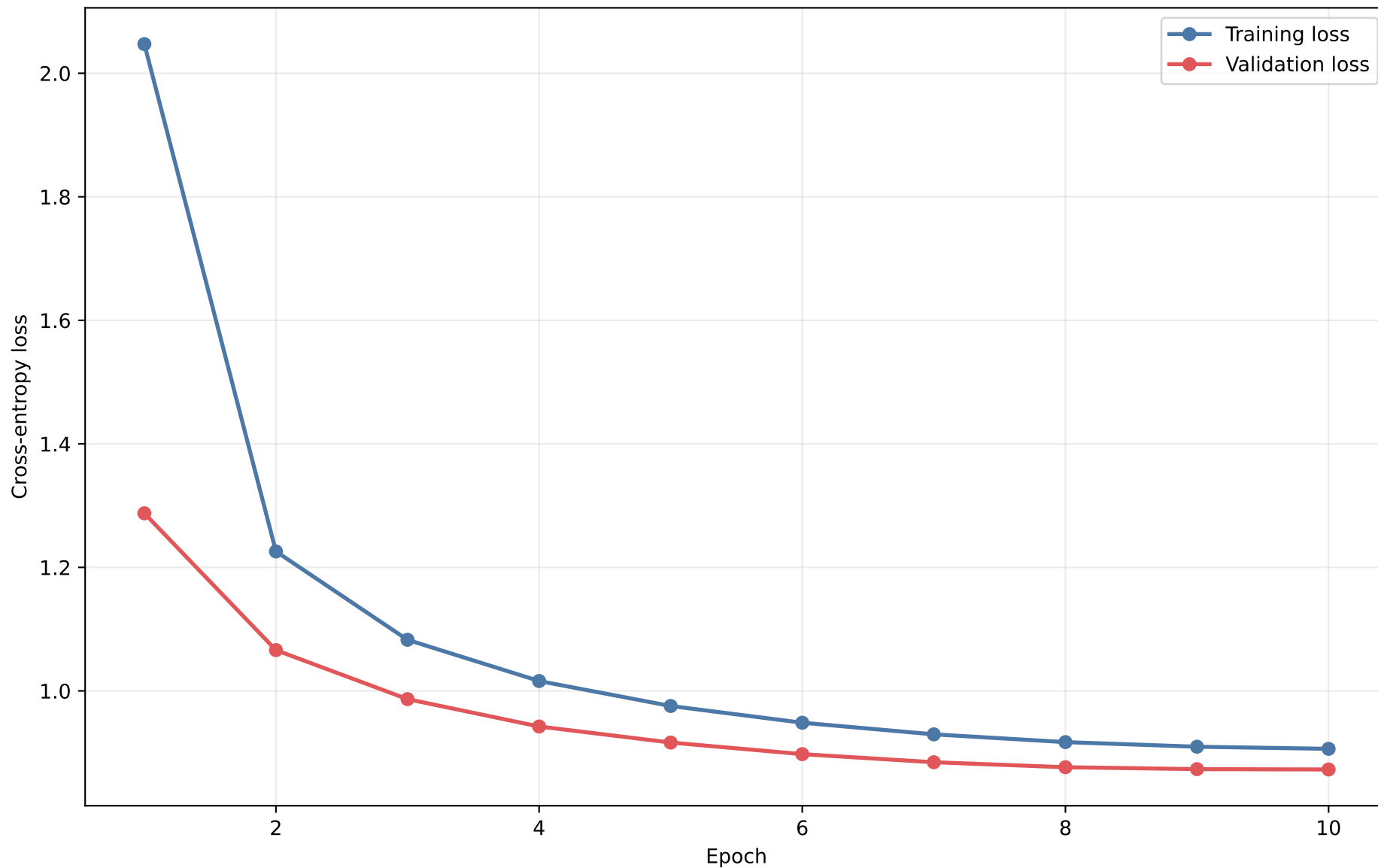
The model contains 3248640 trainable parameters.

The vocabulary was built from the training split only. Unknown validation characters were explicitly mapped to the <UNK> token.

Training and Validation Metrics

Metric	Value
Train CE, dropout ON	0.9062144078662284
Train CE, dropout OFF	0.8433332967150743
Train perplexity	2.324100997948406
Train BPC	1.2166727649873783
Train top-1 accuracy	0.73213515625
Validation CE	0.8727689301891691
Validation perplexity	2.393529201688915
Validation BPC	1.2591394074258802
Validation top-1 accuracy	0.72531796875
Generalization gap, dropout OFF	0.0294356334740948
Generalization gap, dropout ON	-0.0334454776770593

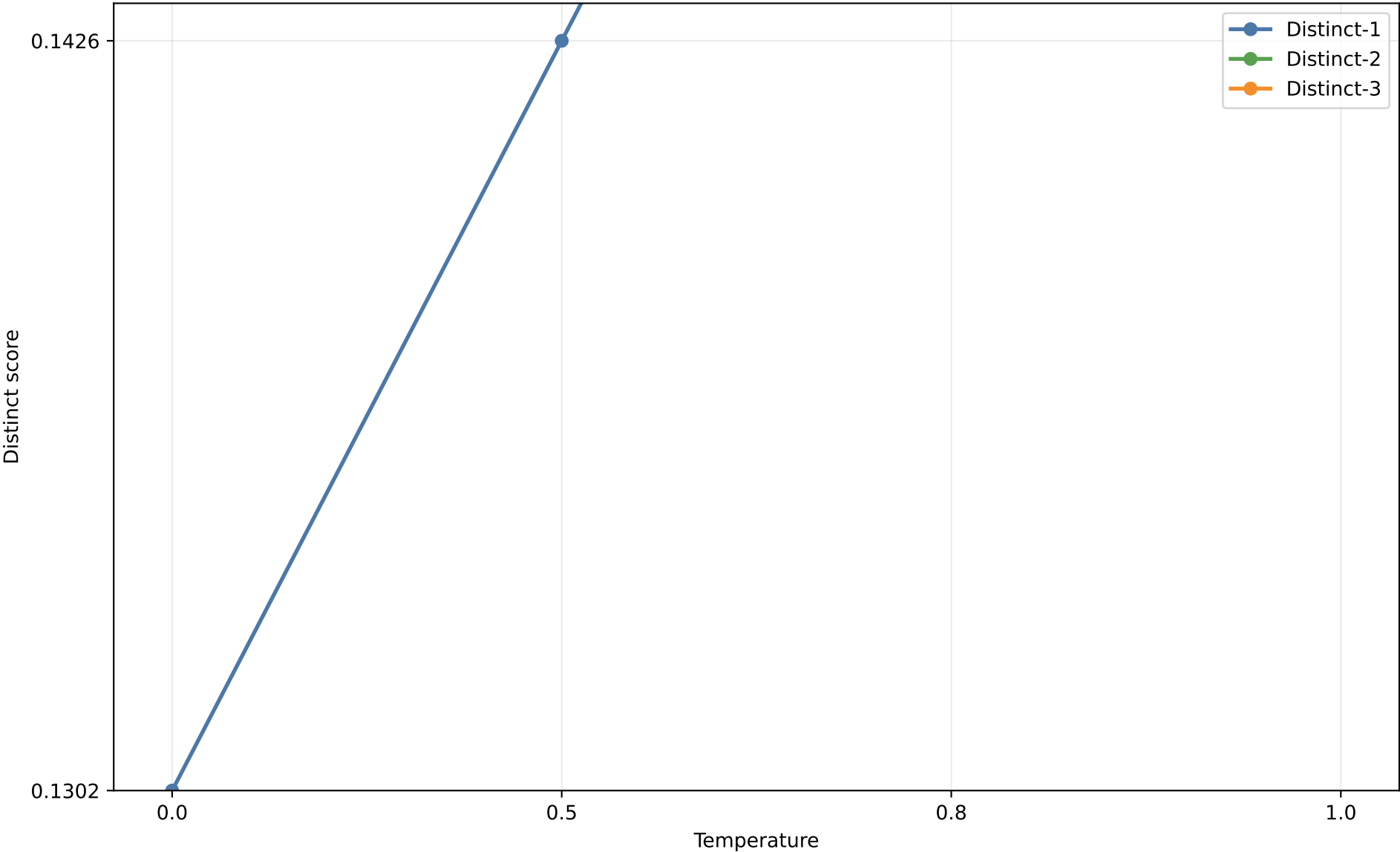
Training and Validation Loss by Epoch



Generation Metrics

Temperature	Distinct-1	Distinct-2	Distinct-3	Repeated 4-gram
0.0	0.1302	0.3882	0.5222	0.4289
0.5	0.1426	0.5235	0.7501	0.1539
0.8	0.1474	0.5615	0.8002	0.1090
1.0	0.1500	0.5752	0.8221	0.0891

Character-Level Generation Diversity



Training Stability

Metric	Value
Mean gradient norm	0.3684396242912351
Maximum gradient norm	1.7694355249404907
Gradient norm standard deviation	0.1008245479209532
Loss spikes	0.0
Non-finite values	0.0

Resource Usage

Metric	Value
GPU	NVIDIA GeForce RTX 4090
PyTorch	2.14.0+cu130
CUDA	13.0
Training time (seconds)	182.5694510936737
Training tokens/second	701103.0554850333
Peak allocated memory (GB)	1.4048843383789062
Peak reserved memory (GB)	1.55078125
Generation characters/second	274.0689248326798

Checkpoint Mapping

Checkpoint	Selection	Validation loss
best_model.pt	lowest validation loss	0.8727689301891691
final_model.pt	Model after final epoch	0.8727689301891691

Saved Artifacts

The Task 1 project contains the following artifacts:

checkpoints/best_model.pt

checkpoints/final_model.pt

outputs/metrics_report.csv

outputs/final_epoch_metrics.csv

outputs/final_training_loss.png

outputs/loss_curves.png

outputs/gradient_norm_and_learning_rate.png

outputs/final_generation_samples.json

outputs/final_generation_metrics.json

outputs/evaluation_summary.json

logs/train_raw.log